

MISFIT: A Self-Supervised Pretraining Toolkit for 3D Medical Imaging

A Preliminary Test Across Three BraTS 2026 Segmentation Tasks

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Introduction

Since its 2015 introduction, U-Net has been cited over 100,000 times, and frameworks like nnU-Net remain the reproducible standard on BraTS and the Medical Segmentation Decathlon. Despite continued architectural innovation — transformers, boundary-aware losses — reported gains over nnU-Net are not always reproducible.

We address this with **MIST**, our modular, end-to-end segmentation framework. Using MIST, we placed **3rd** in both the 2024 BraTS Adult Glioma Post-Treatment challenge and the 2025 BraTS Pre-/Post-Treatment MRI challenge. For BraTS 2026 we enter the three tasks above, each testing a different kind of generalization.

This report asks two questions

- Q1** System-level: does a **MISFIT**-pretrained SwinUNETR-V2 pipeline beat our established pocket nnU-Net baseline?
- Q2** Isolated effect: holding architecture, patch size, and budget fixed, does MAE pretraining itself help over random initialization?

Methods

Table 1 · Dataset sizes and regions evaluated per task				
TASK	COHORT	TRAIN	HELD-OUT	REGIONS
1	Brain metastases	1,196	179	NETC, SNFH, ET, RC
2	Pediatric tumors	294	91	ET, NET, CC, ED
3	BraTS-GoAT	1,351	450	ET, TC, WT

Self-supervised pretraining via MISFIT

MISFIT is our in-development framework for pretraining 3D medical imaging encoders with masked autoencoding (MAE), independent of MIST. Using `misfit_index`, we pooled every training-split MRI volume across all three tasks — including Task 3's 1,137-patient unlabeled cohort — into **16,315** single-channel volumes (T1N/T1C/T2W/T2F indexed separately). A `swinunetr`-base encoder (18.6M params) was trained on 96³ crops with 75% of the volume masked in non-overlapping 16³ cubes, reconstructed via a lightweight transposed-convolution decoder under a per-patch variance-normalized MSE loss (AdamW, lr 1×10⁻⁴, 200 epochs, 4×A100). The pretrained encoder was then transferred into MIST's SwinUNETR-base segmentation architecture.

Three network arms

- Pocket nnU-Net (baseline)** — constant-width (32-filter) conv encoder-decoder, 128³ patch, random init; our returning baseline from prior BraTS challenges.
- SwinUNETR-V2, MISFIT-pretrained** — same architecture, 96³ patch, encoder initialized from MISFIT weights.
- SwinUNETR-V2, from scratch** — identical to the arm above except for random encoder initialization — the matched control.

The two SwinUNETR arms share architecture, patch size, and a 1,000-epoch fine-tuning budget, differing only in encoder initialization — isolating the effect of pretraining. The baseline differs from both in architecture, patch size, learning rate, and budget, so it measures a full pipeline-level contrast instead.

Training & evaluation

5-fold cross-validation, Dice + cross-entropy loss, AdamW, cosine schedule with 20-epoch warmup, bfloat16, GPU-resident DALI augmentation. The baseline trained up to 5,000 epochs/fold (mean 3,107–4,252 actual); both SwinUNETR arms trained exactly 1,000 epochs/fold. Inference used sliding-window Gaussian blending, 8-way flip test-time augmentation, and 5-fold averaging. Arms are compared with MIST's `mist_rank` — a BraTS-style per-patient average rank — and a two-sided Wilcoxon signed-rank test ($\alpha = 0.05$).

Discussion & Conclusion

Held-out validation does not support the pooled MISFIT pipeline beating our established baseline — but the matched control shows MAE pretraining genuinely helps once architecture, patch size, and budget are held fixed.

- Task 1's reversal is explainable.** Task 1 is scored lesion-wise, where a small boundary deficit invisible in per-voxel CV Dice is heavily amplified; a 1,000-epoch fine-tuning budget likely wasn't enough to refine small-lesion boundaries.
- Where pretraining helps most.** Task 2 enhancing tumor Dice improves from 0.518 → 0.670 under cross-validation; on Task 1 held-out, the resection cavity leads on every metric (Dice 0.886 → 0.923) while ET/TC/WT show no advantage — a concentrated, not uniform, gain.
- Transformers are data-hungry without pretraining.** Task 2 (294 cases) is the only task where the from-scratch SwinUNETR fails to converge — and the task where MISFIT initialization helps most.
- Final submission: pocket nnU-Net for all three tasks** — required by data-usage rules barring cross-task pooling, and independently supported by the held-out results themselves.
- This is a preliminary MAE schedule** — 200 epochs against the 800–1,600 convention. Longer pretraining and larger fine-tuning budgets are needed to test whether pretrained transformers can surpass fully converged CNNs on these tasks.

Results

Table 2 · BraTS-style average rank (lower is better), two-sided Wilcoxon p-values						
5-fold cross-validation				Held-out validation		
TASK	BASE	MISFIT	P	BASE	MISFIT	P
1	1.503	1.497	0.62	1.427	1.573	1.5×10⁻⁴
2	1.528	1.472	0.026	1.503	1.497	0.83
3	1.518	1.482	0.0072	1.513	1.487	0.32
Task	MISFIT	Scratch	p	MISFIT	Scratch	p
1	1.439	1.561	2.1×10⁻⁵	1.436	1.564	0.0024
2	1.457	1.543	0.035	1.440	1.560	0.0053
3	1.416	1.584	1.9×10⁻¹³	1.424	1.576	7.4×10⁻⁵

Against the **matched control** (upper vs. lower block, same architecture/patch/budget), MISFIT pretraining wins on all three tasks under both protocols — the isolated pretraining effect is real. Against the **established baseline**, the picture reverses: cross-validation favors the pretrained pipeline on Tasks 2–3, but that advantage vanishes on held-out validation, and the baseline pulls significantly ahead on Task 1 held-out ($p = 1.5\times 10^{-4}$).

Why the CV signal doesn't replicate — the MAE encoder was pretrained on the pooled training split, giving it label-free exposure to images that later formed each cross-validation fold. That asymmetry inflates CV scores for Tasks 2–3; held-out scoring, which excludes the pretraining pool entirely, is unaffected and is why we treat it as the decisive protocol.

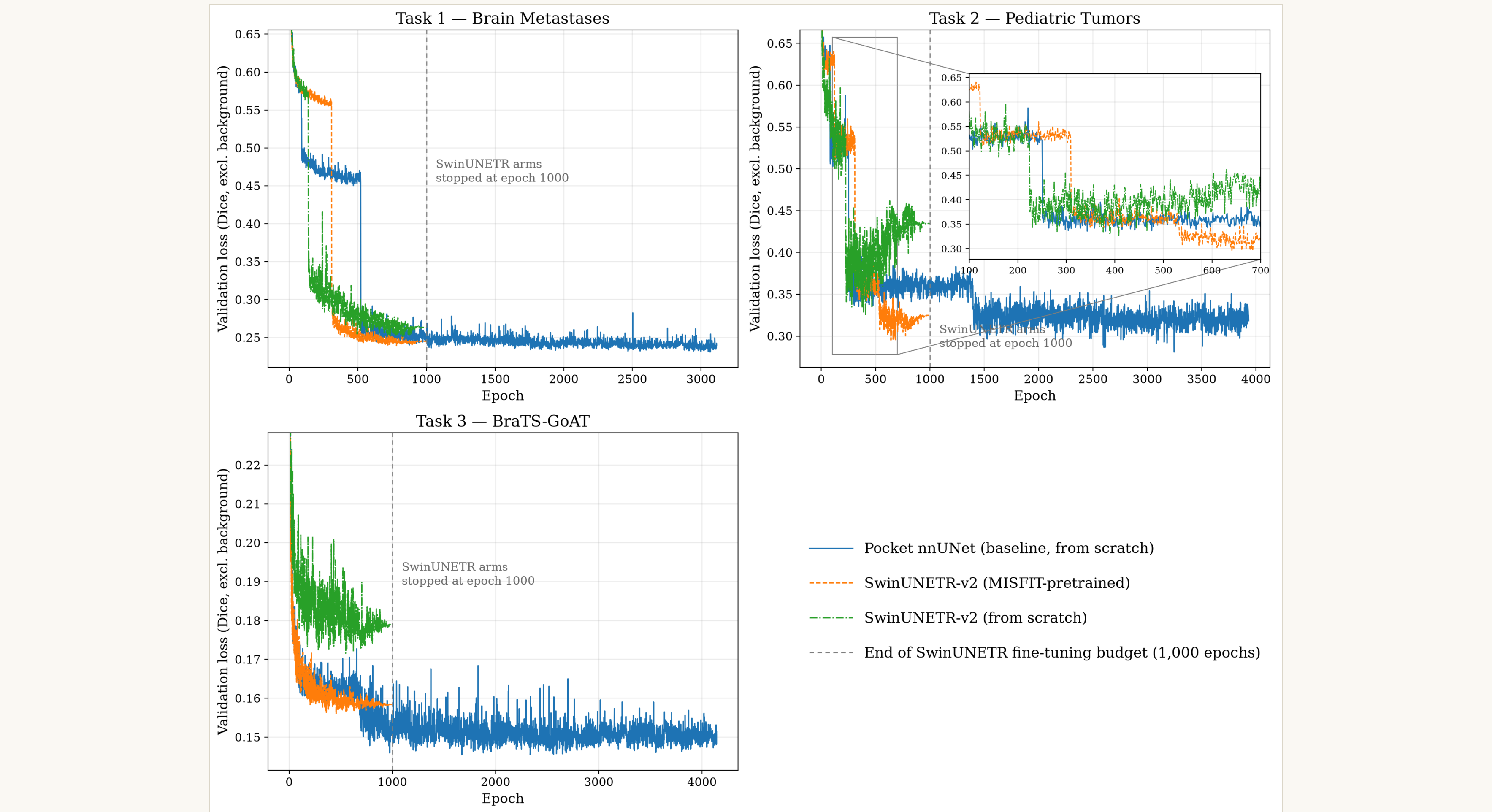


Figure 1. Validation loss (Dice, excl. background) across training. The MISFIT-pretrained arm reaches within 3.5–7.3% of the baseline's fully converged floor inside its 1,000-epoch budget — in far fewer epochs than the baseline needs (means: 3,107 / 3,908 / 4,252). The from-scratch SwinUNETR fails to converge at all on Task 2.

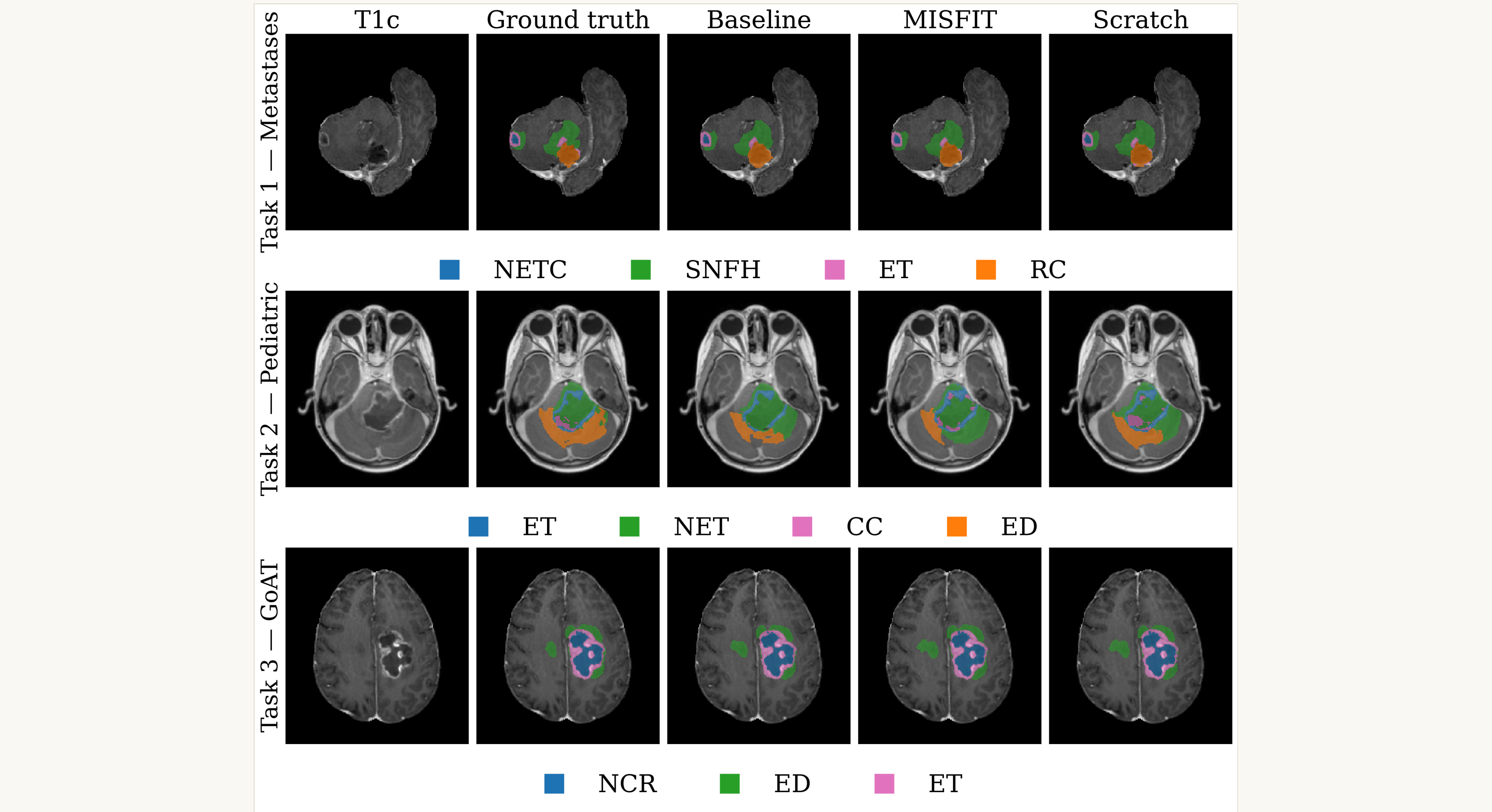


Figure 2. One representative training-cohort case per task: T1c, ground truth, and all three arms. All three agree closely on Tasks 1 and 3; on Task 2 the baseline recovers almost none of the cystic component (pink).

References

[1] Ronneberger, O., et al. U-Net: Convolutional Networks for Biomedical Image Segmentation. *MICCAI*, 2015.

[2] Isensee, F., et al. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 2021.

[3] Isensee, F., et al. nnU-Net Revisited: A Call for Rigorous Validation in 3D Medical Image Segmentation. *arXiv:2404.09556*, 2024.

[4] Hatamizadeh, A., et al. Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images. *MICCAI Brainlesion*, 2021.

[5] He, Y., et al. SwinUNETR-V2: Stronger Swin Transformers with Stagewise Convolutions for 3D Medical Image Segmentation. *MICCAI*, 2023.

[6] He, K., et al. Masked Autoencoders Are Scalable Vision Learners. *CVPR*, 2022.

[7] Tang, Y., et al. Self-Supervised Pre-Training of Swin Transformers for 3D Medical Image Analysis. *CVPR*, 2022.

[8] Celaya, A., et al. PocketNet: A Smaller Neural Network For Medical Image Analysis. *IEEE TMI*, 2023.

[9] Celaya, A., et al. MIST: A Simple and Scalable End-to-End 3D Medical Imaging Segmentation Framework. *arXiv:2407.21343*, 2024.

[10] Celaya, A., et al. A Generalized Surface Loss for Reducing the Hausdorff Distance in Medical Imaging Segmentation. *arXiv:2302.03868*, 2023.

[11] Maier-Hein, L., et al. Metrics Reloaded: Recommendations for Image Analysis Validation. *Nature Methods*, 2024.

[12] Moawad, A.W., et al. The Brain Tumor Segmentation (BraTS-METS) Challenge 2023. *arXiv:2306.00838*, 2023.

[13] Kazerooni, A.F., et al. The Brain Tumor Segmentation in Pediatrics (BraTS-PEDs) Challenge. *arXiv:2404.15009*, 2024.

[14] Menze, B.H., et al. The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS). *IEEE TMI*, 2015.

[15] Wu, L., et al. VoCo: A Simple-yet-Effective Volume Contrastive Learning Framework for 3D Medical Image Analysis. *CVPR*, 2024.

[16] Loshchilov, I., Hutter, F. Decoupled Weight Decay Regularization. *arXiv:1711.05101*, 2017.